Misallocation in Input Market and Labor Reallocation: Evidence from India

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Overview

- State provision of electricity is widespread across countries
- Political and distributional concerns play important role in electricity pricing and provision
- Efficiency dictates that prices should equal the social marginal cost (SMC) of providing electricity
- **State provision $\rightarrow$ preferential pricing for certain fractions of consumers** (Mahadevan (2021))

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- Indian Context: Low electricity prices for agricultural and residential consumers, high prices for industrial consumers
- Electricity price for Industrial consumers - Electricity prices for farmers/residential consumers= Cross-Subsidy
- **Does this lead to misallocation of labor across sectors?**

Anecdotal Evidence - Political Rate Setting

Congress Promises Free Electricity Scheme For Farmers If Voted To Power In MP

Jolted by power row, Cong says 24x7 free electricity part of poll manifesto

KCR promises free electricity for farmers 'if non-BJP govt comes to power'

Amid the debate over the freebies, Telangana Chief Minister K Chandrashekar Rao announced to provide a "free" power supply to all farmers "if a non-BJP government comes to power"

How Agriculture Consumes 23% Of India's Electricity & Picks 7% Of Tab

Anecdotal Evidence - Political Rate Setting

Electricity overcharging: Cross-subsidies cost businesses Rs 75k crore in FY19

Nitin Raut: Doing away with cross-subsidy in power sector will adversely affect farmers, poor

Silent on free farm power, Montek panel says cross-subsidy is hurting industry

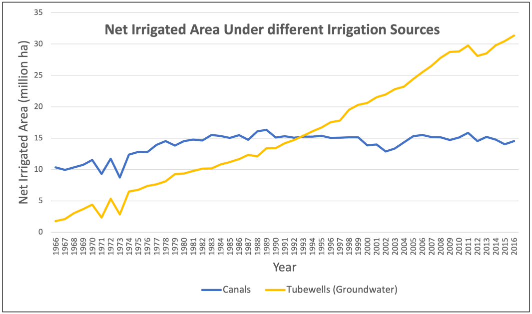
The Montek Singh panel's final report has underlined that "cost and quality of power is a critical factor affecting the competitiveness of industry in the state

Cross-subsidy on power should go: Niti Aayog

3 min read • 28 Jun 2017, 09:37 AM IST

Overview

- Electricity used for running tubewells, which pump groundwater for irrigation
- Government of India in the 1960s began to subsidize a number of key agricultural inputs, **including electricity for groundwater irrigation.**
- Politicians took to using electricity pricing as a campaign tool, heavily subsidizing it for farmers and residential consumers
- Area under tubewell irrigation has boomed as a consequence of cheap electricity for agriculture; 20% area under agricultural holdings irrigated by tubewells



Overview

- The industrial sector uses electricity for operating capital
- In a 2006 World Bank survey of Indian manufacturing firms, more than 36% mentioned electricity as the biggest constraint to their operations (Abeberese (2017), Allcott, Collard-Wexler and O'Connell (2016))
- Depending on whether labor and capital are complements or substitutes, higher electricity prices for manufacturing may lead to higher or lower labor
- Industrial and Commercial consumers cross-subsidize residential and agricultural consumers by paying high prices

Research Question

- Do distortions in factor markets lead to misallocation of labor across sectors(agricultural/manufacturing)?

This project: Distortion in electricity tariffs across agricultural and industrial consumers in India ('cross-subsidy')

Data

Employment

- **NSS Employment and Unemployment Surveys:** Sector of employment; industry of employment (equivalent of SIC codes)
  - 2004, 2005, 2006, 2008, 2010, 2012
  - Individual-level
- **NSS Periodic Labor Force Surveys:** Sector of employment; industry of employment (equivalent of SIC codes)
  - 2017, 2018, 2019, 2020
  - Individual-level

Data

Electricity Prices

- **Central Electricity Authority of India**'s *Electricity Tariff & Duty and Average Rates of Electricity Supply in India* reports
  - Annually (only available from 2011-2019 online)
  - State-level
- **Power Finance Corporation of India**'s *Performance of State Power Utilities* reports
  - Annually (2002-2019)
  - Reports revenues and consumption at the utility-year-consumer level

Coal Prices

- **Ministry of Coal's** *Coal Directory of India*
  - Country-Year level (2002-2019)

Data Background

- Most farmers' tubewells are unmetered i.e., cannot record their actual electricity consumption
- Cannot assign a unit price (Rs./Unit) in this case
- Most farmers face flat tariffs i.e., they pay a flat monthly price irrespective of actual use, depending on the capacity of the tubewell
- To prevent overuse, supply is rationed
- Prices set by the electricity utilities, and state govt. will provide subsidies for agricultural consumers on top of this
- Subsidy may or may not reflect in reported tariffs

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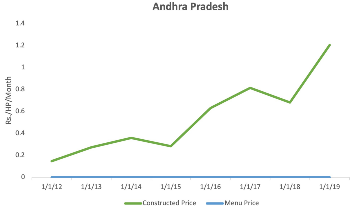
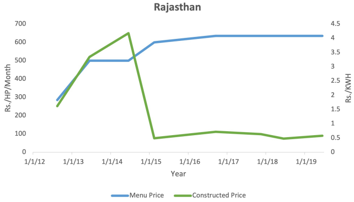
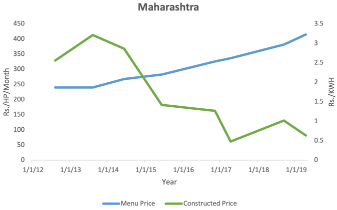
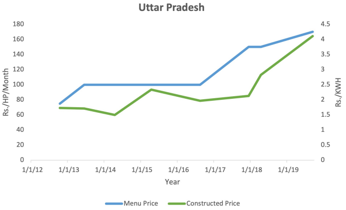
To bring everything to price per unit measure and account for subsidies, we construct average price for agricultural and industrial consumers using electricity revenue and sales data PFC

Comparison of Reported Agricultural Price and Constructed Agricultural Price

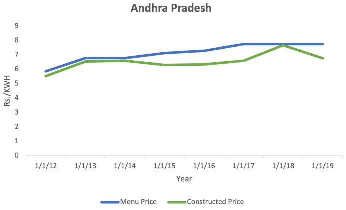


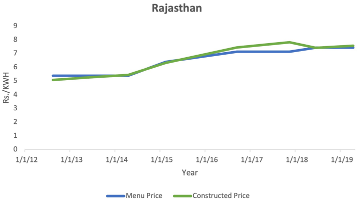


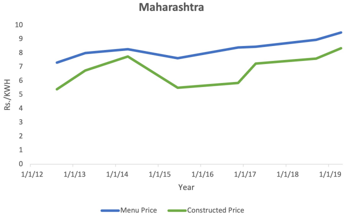


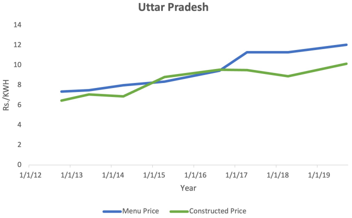


Comparison of Reported Industrial Price and Constructed Industrial Price

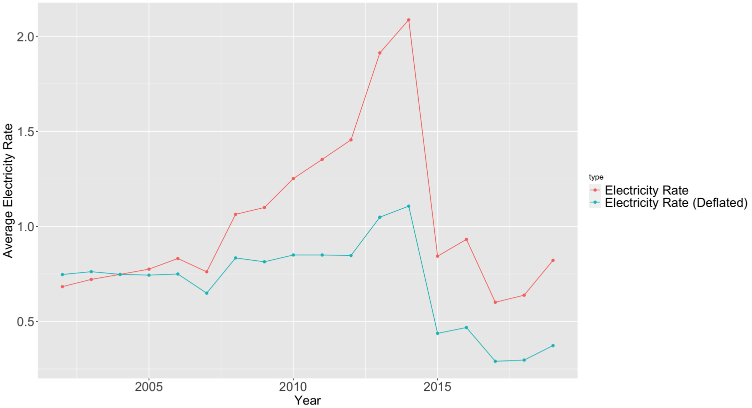




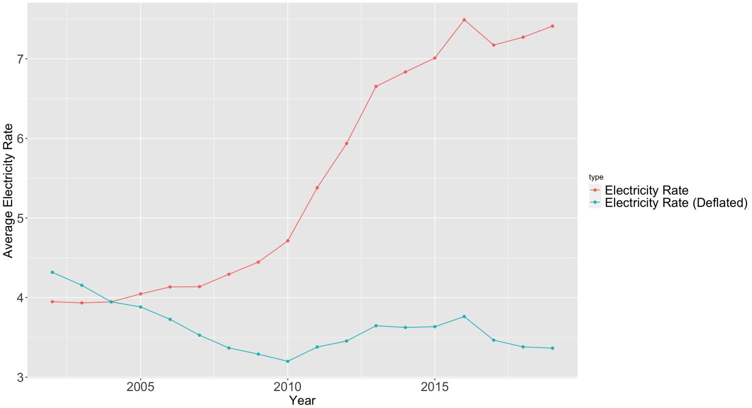




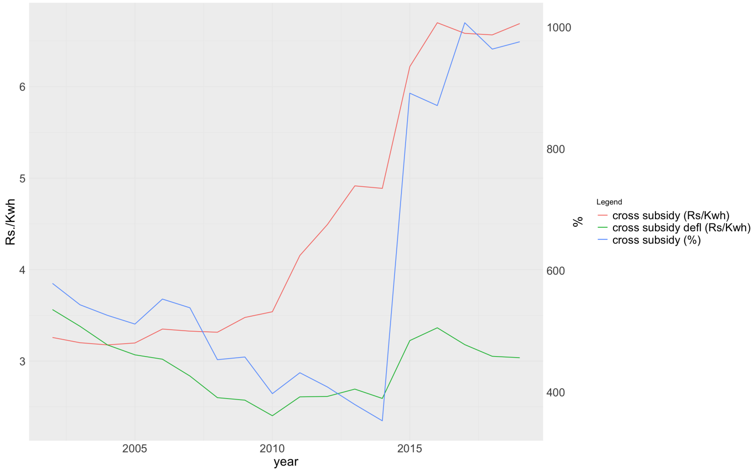
Agricultural Price Trend



Industrial Price Trend



Cross-Subsidy Trend



▸ CEA PFC Agri ▸ CEA PFC Industrial ▸ CEA PFC Industrial

Empirical Strategy 1: OLS Specification

$$LaborWedge_{st} = \beta(ElecPriceWedge)_{st} + X_{st} + \epsilon_{st}$$

- *LaborWedge*: Share(Agri)-Share(Manuf)
- *PriceWedge* : p(Manuf)-p(Agri)
- j : Consumer Category (Agricultural/ Industrial)
- s : State
- t : Year
- X : Controls

OLS Results

Table 1: Impact of Price Wedge ($p(Manuf) - p(Agri)$) on Employment Wedge ($Share(Agri) - Share(Manuf)$)

	(1)	(2)	(3)	(4)
CS Rupees	0.0640	-0.0426	0.0121	0.0545
	(0.1690)	(0.2545)	(0.0787)	(0.0446)
Obs	176	176	176	176
Region FE	No	Yes	No	No
State FE	No	No	Yes	No
Year FE	No	Yes	Yes	No
Region-Year FE	No	No	No	Yes
State Trend	No	No	No	No

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Endogeneity Concerns

- Electricity prices endogenous
- **Use a shift-share IV strategy, coal price $\times$ baseline thermal share of generation in state** (Abeberese (2017))

Empirical Strategy 2: Shift-Share IV

- Coal occupies lion's share in electricity generation in India
- Changes in coal prices may pass through to final electricity rates

- 83% of coal in India produced by a public company viz. Coal India Limited (henceforth CIL)
- Coal prices set nationally by CIL, revised time to time for offsetting inflationary pressures on their costs, increases in wage bill etc
- Separate prices for power utilities and other consumers

Coal Prices

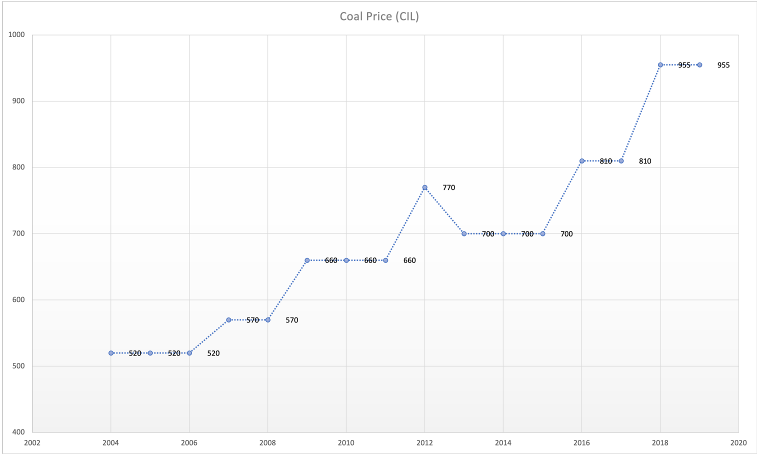


Figure 1: Coal Prices

IV Results - Wedge (Rupees)

Table 2: Impact of Price Wedge ($p(Manuf) - p(Agri)$) on Employment Wedge ($Share(Agri) - Share(Manuf)$)

	(1)	(2)	(3)	(4)
Panel A: First Stage				
Thermal Share (2002) × Coal Price	0.0027**	0.0042***	0.0018	0.0051***
	(0.0014)	(0.0013)	(0.0035)	(0.0014)
Kleibergen-Paap Wald F statistic	4.07	9.41	0.53	13.31
Panel B: Second Stage				
CS Rupees	-0.0384	-0.0080	0.0018	0.0547**
	(0.090)	(0.059)	(0.104)	(0.0259)
Obs	176	176	176	176
Region FE	No	Yes	No	Yes
State FE	No	No	Yes	No
Year FE	No	Yes	Yes	No
Region-Year FE	No	No	No	No
State Trend	No	No	No	Yes

Conclusion

- **Main Finding:** IV estimates show that a higher electricity price wedge is associated with a higher employment wedge
  - A Rs. 1/kWh increase in electricity price cross-subsidy increases labor share wedge by 5.5 percentage points
  - Effect is robust to region-year FE and state trends

- **Contribution:**
  - Highlights electricity pricing distortions as a potentially overlooked driver of labor misallocation in developing countries
  - Politically-motivated cross-subsidies may hinder structural transformation by raising costs for industry while artificially sustaining agricultural employment
  - Complements existing literature on electricity *access* and *reliability* by focusing on *pricing* distortions

Example: Tariff Order

ELECTRICITY TARIFF IN JHARKHAND w.e.f. 01.01.2016

Category No.	Category of Service	Fixed Charge	Energy Charge Paise/kWh
DS	**DOMESTIC SERVICE**		
DS-I(a)	Kutir Jyoti for load upto 100 Watts for rural areas		
	0-100 kWh/Month (metered)	15 Rs/Month	120
	Kutir Jyoti (Unmetered)	40 Rs/Month	Nil
DS-I(b)	For other consumers in rural areas with connected load upto 2 KW (metered) & not coveed in DS-II		
	0 - 200 kwh/month	27 Rs/Month	150
	Above 200 kWh/Month	27 Rs/Month	160
	For other consumers in rural areas connected load upto 2 KW (Unmetered)	110 Rs/Month	Nil
DS- II	For urban areas covered by Notified Area Committee/ Municipality/ Municipal Corporation/ All Disstt.Towns/ All Sub-Divisional Towns / All Block Headquarters/ Industrial Areas/ Contiguous sub-urban areas all market places urban or rural and for connecting load upto 4 KW		
	0-200 kWh/Month	43 Rs/Month	260
	Above 200 kWh/Month	65 Rs/Month	310
DS- III	For load exceeding 4 KW and upto 85 KW	110 Rs/Month	320
DS-HT	Supply to housing complex/ multistoreyed buildings at 11 KV & load above 85.044 KW	80 Rs/KVA/Month	280
NDS	**NON-DOMESTIC SERVICE**		
NDS-I	For rural areas connected load upto 2 KW (metered) - All Consumption	32 Rs/Month	190
	For rural areas connected load upto 2 KW (Unmetered)	190 Rs/Month upto 1 kW &	Nil
		70 Rs for every additional kW	
NDS - II	For urban areas for connected loads upto 85.044 KW covered by Notified Area Committee/ Municipality/ Municipal Corporation/ All Disstt.and Sub-Divisional Towns /All Block Headquarters/ Industrial Areas and Contiguous Sub-urban areas. Also applicable to co		
	All consumption	190 Rs./kW/Month	565
NDS - III	**ADVERTISING & HOARDINGS**		
	For Single Phase for load upto 2 KW &Three Phase for load exceeding 2 KW.	165 Rs./Month	650
LTIS	**LOW TENSION INDUSTRIAL & MEDIUM POWER SERVICE ***		
	Installation Based Tariff upto 114 HP	140 Rs/HP/Month	530
	Demand Based Tariff upto 100 KVA	255 Rs/KVA/Month	530
IAS	**IRRIGATION & AGRICULTURE SERVICE**		
IAS - I	For private tubewells & private lift irrigation schemes (metered)	Nil	65
	For private tubewells & private lift irrigation schemes (Unmetered)	75 Rs/HP/Month	Nil
IAS - II	For State tubewells & State lift irrigation schemes (metered)	Nil	110
	For State tubewells & State lift irrigation schemes (Unmetered)	300 Rs/HP/Month	Nil
HTS	**HIGH TENSION VOLTAGE SUPPLY SERVICE**		
	Consumers having contract demand above 100 KVA (AT 11/ 33/ 132 kV & above)	255 Rs/KVA/Month	585
	TIME OF THE DAY (TOD) TARIFF FOR HTS CONSUMERS		
	PEAK HOURS (06.00 AM to 10.00 AM & 06.00 PM to 10.00 PM)	120% of normal rate of energy charge	
	OFF-PEAK HOURS (10.00 PM to 06.00 AM)	85% of normal rate of energy charge	

return

Example: Tariff Order

IAS	**IRRIGATION & AGRICULTURE SERVICE**		
IAS - I	For private tubewells & private lift irrigation schemes (metered)	Nil	65
	For private tubewells & private lift irrigation schemes (Unmetered)	75 Rs/HP/Month	Nil

LTIS	**LOW TENSION INDUSTRIAL & MEDIUM POWER SERVICE ***		
	Installation Based Tariff upto 114 HP	140 Rs/HP/Month	530
	Demand Based Tariff upto 100 KVA	255 Rs/KVA/Month	530
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	OFF-PEAK HOURS (10.00 PM to 06.00 AM)	85% of normal rate of energy charge	
HTSS	**HT SPECIAL SERVICE FOR INDUCTION / ARC FURNACE**		
	Consumers having contract demand of 300 KVA and more (AT 11/ 33/ 132 kV & above))	440 Rs/KVA/Month	350

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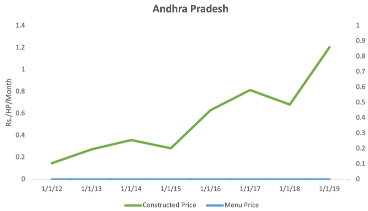
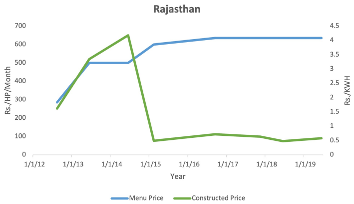
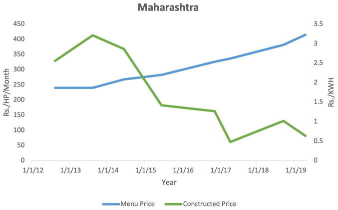
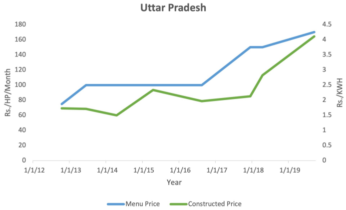
Example: Comparison of PFC and CEA Agri prices









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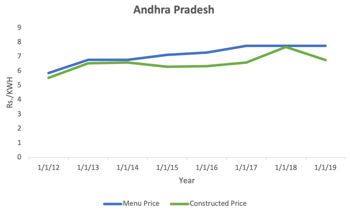
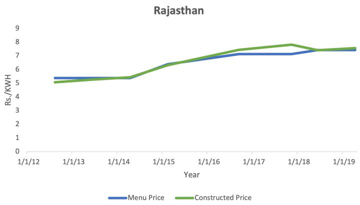
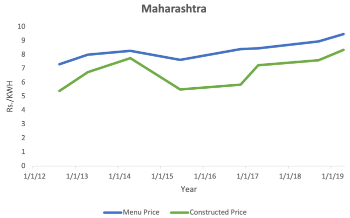
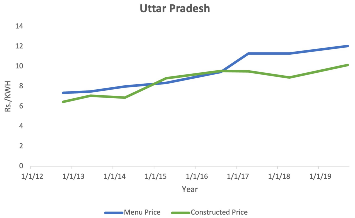
Example: Comparison of PFC and CEA Industrial prices









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Electricity Price Construction

Sales to different consumer categories (Mkwh)

			2013-14		2014-15		2015-16	
Region	**State**	**Utility**	**Agricultural**	**Total Industrial**	**Agricultural**	**Total Industrial**	**Agricultural**	**Total Industrial**
Eastern	**Bihar**	NBPDCL	105	442	103	493	131	566
		SBPDCL	217	1,393	210	1,361	224	1,616
	Jharkhand	JSEB	60	1,897				
		JBVNL	19	638	87	2,428	112	3,623
	Odisha	CESU	46	1,764	56	1,747	47	1,433
		NESCO	49	1,612	52	1,557		
		SESCO	30	395	32	401		
		WESCO	61	2,510	83	2,584		
		NESCO Utility					60	1,754
		SESCO Utility					42	384
		WESCO Utility					115	2,228
	Sikkim	Sikkim PD		110	0	112		128
	West Bengal	WBSEDCL	1,183	7,087	1,492	7,154	1,524	6,762

Revenue from different consumer categories (Rs Crores)

			2013-14		2014-15		2015-16	
Region	**State**	**Utility**	**Agricultural**	**Total Industrial**	**Agricultural**	**Total Industrial**	**Agricultural**	**Total Industrial**
Eastern	**Bihar**	NBPDCL	45	301	48	336	57	414
		SBPDCL	50	851	55	847	64	1,039
	Jharkhand	JSEB	4	1,092				
		JBVNL	1	346	6	1,451	6	1,586
	Odisha	CESU	6	1,007	7	852	8	762
		NESCO	6	954	8	943		
		SESCO	6	229	0	237		
		WESCO	10	1,545	9	1,580		
		NESCO Utility					10	1,023
		SESCO Utility					7	230
		WESCO Utility					17	1,394
	Sikkim	Sikkim PD		61		68		76
	West Bengal	WBSEDCL	350	5,018	578	5,530	620	5,068

return

Electricity Price Construction

State	Utility	Year	Agri_price	Ind_price
andhra pradesh	APCPDCL	2002	0.573839662	4.506486
andhra pradesh	APEPDCL	2002	0.321100917	2.931206
andhra pradesh	APNPDCL	2002	0.654320988	4.294416
andhra pradesh	APSPDCL	2002	0.879120879	4.240212
andhra pradesh	APCPDCL	2003	0.300838402	4.264740
andhra pradesh	APEPDCL	2003	0.405300078	3.683394
andhra pradesh	APNPDCL	2003	0.328386486	4.314869
andhra pradesh	APSPDCL	2003	0.478971963	4.205318
andhra pradesh	APCPDCL	2004	0.349475786	4.172715
andhra pradesh	APEPDCL	2004	0.395284327	2.987826
andhra pradesh	APNPDCL	2004	0.340030912	4.305816
andhra pradesh	APSPDCL	2004	0.342304677	3.994681
andhra pradesh	APCPDCL	2005	0.240430244	3.921089
andhra pradesh	APEPDCL	2005	0.124908156	3.131684
andhra pradesh	APNPDCL	2005	0.070357908	4.072693
andhra pradesh	APSPDCL	2005	0.053560176	3.868313
andhra pradesh	APCPDCL	2006	0.089957035	3.801098
andhra pradesh	APEPDCL	2006	0.128847530	3.235837
andhra pradesh	APNPDCL	2006	0.057727631	3.868170
andhra pradesh	APSPDCL	2006	0.035059331	3.853282
andhra pradesh	APCPDCL	2007	0.116136919	3.212101
andhra pradesh	APEPDCL	2007	0.132645542	3.669841

return

Model: Simpler Problem 2 (with mods)

- Let agent i have productivities z_f, z_m in farming and manufacturing respectively.
- Profit function in farming: $\pi_f = \max_{\{\bar{L}_f, E_f\}} PE_f^{\alpha} \bar{L_f}^{1-\alpha} - w_f \bar{L_f} - p_f^e E_f$
- $w_f = (1-\alpha)P^{\frac{1}{1-\alpha}}(\frac{\alpha}{p_f^e})^{\frac{\alpha}{1-\alpha}}$
- Labor earnings of agent who has productivity z_f in farming is $z_f w_f$.

- Profit function in manufacturing: $\pi_m = \max_{\{\bar{L}_m, E_m\}} E_m^{\beta} \bar{L_m}^{1-\beta} - w_m \bar{L_m} - p_m^e E_m$
- $w_m = (1-\beta)(\frac{\beta}{p_m^e})^{\frac{\beta}{1-\beta}}$
- Labor earnings of agent who has productivity z_m in farming is $z_m w_m$.

- Agent chooses to be in agriculture if $z_f w_f \geq z_m w_m$. Let Ω_f, Ω_m be the sets of workers choosing to be
- The total number of workers in each sector is as follows:

Model: Simpler Problem 2 (with mods)

- Following **?**, let z_f and z_m be drawn independently from Frechet distributions as follows:

$$G(z_f) = e^{-z_f^{-\theta}}; \quad G(z_m) = e^{-z_m^{-\theta}}$$

- The parameter θ controls the dispersion of individual productivity in the sectors
- The share of workers in agriculture and manufacturing are as follows:

$$S_f = \frac{\Phi_f^\theta}{\Phi_f^\theta + \Phi_m^\theta}; \;\; S_m = \frac{\Phi_m^\theta}{\Phi_m^\theta + \Phi_m^\theta}$$

$$\Phi_f = (1-\alpha)P^{\frac{1}{1-\alpha}}(\frac{\alpha}{p_f^e})^{\frac{\alpha}{1-\alpha}}$$

$$\Phi_m = (1-\beta)(\frac{\beta}{p_m^e})^{\frac{\beta}{1-\beta}}$$

Model: Simpler Problem 2 (with mods)

- We need to identify $\alpha, \beta, \theta_f, \theta_m, P$.
- In data, we only see a worker in one sector with following condition satisfied $z_f w_f \geq z_m w_m$.
- Given the technology is CRS: $\alpha = \frac{p_f^e E}{Y_f} \quad \beta = \frac{p_m^e E_m}{Y_m}$
- θ is disciplined by mean and variance of wages in the data.
- **Most important** What matters for the movement of a manufacturing worker to agriculture is his unobserved productivity, z_f. We need to have an idea of this through a joint distribution of z_f and z_m, so that we can make precise statements about whether he is going to move or not when conditions change in the manufacturing sector, for example, due to changes in electricity prices.

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References I

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